

PAOS AI_Workflow

Coding Principles Benchmark

Grade A - 181/200 (91%)

Category	Score	Max	%	Grade
1. Object-Oriented Programming	31/36	36	86%	B
2. Data Structures	15/18	18	83%	B
3. Graph Theory & Discrete Math	11/12	12	91%	A
4. Linear Algebra	7/8	8	87%	A
5. Digital Logic & Design	8/8	8	100%	A
6. Numerical Analysis	7/8	8	87%	A
7. System Analysis & Design	14/14	14	100%	A
8. Database (File-as-Database)	14/16	16	87%	B
9. Calculus I-III	7/10	10	70%	B
10. Human Error & UX/UI	20/20	20	100%	A
11. Pipeline-Specific	29/20	20	145%	A
12. Code Quality	18/20	20	90%	A
TOTAL	181/200	200	91%	A

STRENGTHS (100% Scores)

- Digital Logic: combinatorial permissions, sequential cascade, crash recovery
- System Analysis: clean layering, low coupling, high cohesion, queue separation
- UX/UI: full Nielsen heuristics, 390px-4K responsive, scroll-wheel sliders
- Pipeline: naming convention, cascade, retry/skip, enrichment, smart animation

TOP IMPROVEMENTS NEEDED

1. Replace `any` types (Q92)

5 files use `: any`. Replace with proper interfaces.

2. Add cycle detection messages (Q21)

Kahn's algo silently truncates cycles. Add explicit error.

3. File read caching (Q56)

Add in-memory cache with TTL for META.json, queue.json.

4. DAG validation on every read (Q27)

Validate node refs, no duplicate IDs, required fields.

5. Velocity tracking & trends (Q62/64)

Tasks/sec rate, ETA from moving average.

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6. Similarity metrics (Q37)

Cosine similarity for prompt matching, vector profiles.

7. React.memo on children (Q100)

Memo wrappers on Canvas children, FileTreeExplorer.

8. Reduce prop drilling (Q4)

Pass narrow subsets instead of full pipeline objects.

OVERALL ASSESSMENT

The PAOS AI_Workflow codebase demonstrates strong adherence to computer science and mathematics principles. The architecture is cleanly layered, data flows follow predictable paths, and the pipeline system correctly implements DAG theory with deterministic execution ordering. The codebase excels in UX/UI design and digital logic with perfect scores in error recovery, responsive design, and state management. Areas for growth include TypeScript strictness (eliminating `any`), adding proper caching, and implementing analytical features like velocity tracking and similarity metrics for template suggestions. The Grade A rating reflects production-ready quality with minor, targeted improvements needed.